

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
2.	(a)	(i)		Endangered – (existing in such small numbers that it is) {in danger of/ at risk of/ close to} {becoming extinct/ dying out/ owtte} / very few (individuals) remaining and Extinct - there are no more individuals of that species alive (anywhere in the world)/ species {has died out/ owtte}/ none of a species remaining (1)	1			1		
		(ii)		Any two (x1) from • Pollution (1) • {Hunting/ fishing} (1) • Introduction/ competition of alien species (1) • Competition from domestic animals (1) • Natural selection (1) • Global warming/ climate change (1)	2			2		
	(b)	(i)		Secondary succession (1)	1			1		
		(ii)		Any three (x1) from A. Presence of {soil/ spores/ seeds/ owtte} B. Pioneer species are the first to return C. {presence alters conditions/ owtte} allowing other species to grow (1) D. Seres/ seral stages + the sequential progression of species to form intermediate communities/ or description of (1) E. Climax community + remain {stable / in equilibrium/ owtte} / final stage of succession (1)	3			3		

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	(c)	(i)		{Population/ gene pool} in fragment would be small (HW would not apply) (1) Genetic drift will cause changes in allele frequencies/ ORA (1)		1	1	2		
		(ii)		• Most risk- Q P R -least risk (1) • Correct reference to {ease/ difficulty} of {gene flow/ reproducing with main population} (1) • R is {attached/ has corridor} (1) • Q is {further away than P/ most isolated/ furthest away}/ ORA (1) Alternative: • Most risk- PQ R -least risk (1) • Correct reference to {ease/ difficulty} of {gene flow/ reproducing with main population} (1) • R is attached/ has corridor (1) • P was separated before the others (1)		3	1	4		
	(d)	(i)		41.2 % = 2 marks If incorrect award 1 mark for any of 41.22 $(300^2 - 230^2) / 300^2 * 100$ $37100 / 90000 * 100$		2		2	2	
		(ii)		Smaller the patch the greater the {percentage/ area} influenced by edge effects/ ORA (1) Any two (x1) from light (intensity/ wavelength) (1) temperature (1) humidity (1) wind/ air movement (1)		3		3		
				Question 2 total	7	9	2	18	2	0